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Distribution and population trends of the European Roller in pseudo-steppe areas of Portugal: results from a census in sixteen SPAs and IBAs

Distribuição e tendências populacionais do rolieiro em áreas de pseudo-estepe em Portugal: resultados de um censo em dezasseis ZPE's e IBA's

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ABSTRACT - The Roller (*Coracias garrulus*) is one of the seventeen Portuguese breeding birds classified as Critically Endangered. Nonetheless, despite its conservation status, it has received very little attention from scientists and conservationists in Portugal. Its population size is unknown and there is limited information on the distribution and population trends of this species. Therefore, a large-scale assessment of the Roller population was carried out in 2009 in sixteen Special Protection Areas (SPAs) and Important Bird Areas (IBAs) with cereal-steppic characteristics in the Alentejo and Beira Baixa regions. During the breeding season, 1570 human made structures were surveyed in an area of more than 320 000 hectares in order to assess the occurrence and abundance of rollers and to study the selection of structures by breeding pairs. Overall, the Roller population was estimated at 57 to 62 breeding pairs, distributed in only six of the sixteen prospected areas. Castro Verde was the most important area with 52-55 pairs, followed by residual populations in Vila Fernando (2-3 pairs), Évora, S. Vicente and Cuba (1 pair) and Moura-Mourão-Barrancos (0-1 pair). Our results confirm the scarcity and highly fragmented distribution of the species, suggesting a general negative trend of the Portuguese population. The characterization of building structures in the surveyed areas suggests that the presence of the species might be constrained by the lack of suitable cavities for nesting. Rollers positively selected adobe-built structures surrounded by fallows. Only in Castro Verde the species is relatively abundant and is

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likely to be increasing as a result of the high availability of suitable breeding sites and high quality habitat. This study highlights the need for urgent intervention in order to protect this endangered species. The maintenance and restoration of the traditional extensive cereal cultivation in the surveyed SPAs and IBAs (threatened by agriculture intensification) and the provisioning of artificial nest-sites are likely the most effective measures to guarantee the species conservation.

RESUMO - O Rolieiro (*Coracias garrulus*) é uma das dezassete espécies de aves nidificantes em Portugal com estatuto de Criticamente Ameaçada. Apesar do seu estatuto de conservação desfavorável, o rolieiro tem recebido muito pouca atenção quer do ponto de vista científico quer do ponto de vista conservacionista. O tamanho da população reprodutora não é conhecido e a informação sobre a sua distribuição e tendência populacional é escassa. Em 2009, com o objectivo de aumentar o conhecimento sobre a espécie em Portugal, realizou-se um censo nacional em dezasseis Zonas de Protecção Especial (ZPE) e Áreas Importantes para as Aves (IBA) com habitat estepário no Baixo Alentejo e Beira Baixa. Ao longo da época reprodutora, foram visitadas 1570 estruturas (casas, barracões agrícolas, igrejas, ruínas, etc) numa área superior a 320 000 hectares com o objectivo de estimar o número de casais de rolieiro e de estudar a selecção do habitat de nidificação. A população de rolieiro foi estimada em 57 a 62 casais, distribuídos por apenas seis das dezasseis ZPEs/IBAs prospectadas. A maioria da população encontra-se na região de Castro Verde (52-55 casais), seguindo-se as áreas de Vila Fernando (2-3 casais), Évora, S. Vicente e Cuba (1 casal) e Moura-Mourão-Barrancos (0-1 casais). Os resultados deste trabalho realçam a reduzida dimensão da população de rolieiro e a sua distribuição altamente fragmentada, sugerindo ainda um recente decréscimo populacional. Os rolieiros seleccionaram como local de nidificação estruturas de adobe rodeadas por pousios, sugerindo que a espécie poderá estar condicionada pela falta de locais adequados à nidificação. Apenas na região de Castro Verde a espécie é razoavelmente abundante e a sua população estará mesmo a aumentar, provavelmente em consequência do elevado número de estruturas adequadas à nidificação e elevada disponibilidade de habitat. Este estudo realça a urgência na implementação de medidas de conservação dirigidas ao rolieiro. A manutenção e gestão do cultivo extensivo de cereais de forma tradicional nas ZPEs e IBAs visitadas (ameaçadas pela intensificação agrícola) e a disponibilização de ninhos artificiais constituem as medidas mais eficazes para garantir a conservação desta espécies.

The Roller (*Coracias garrulus*) is a migratory visitor to the Palaearctic region, breeding from north-west Africa, the Iberian Peninsula and Mediterranean countries to western Himalayas (Cramp & Simmons 1988). Rollers are obligate cavity-nesters, using abandoned woodpecker cavities in trees, fissured rocks or sandy banks to nest (Cramp & Simmons 1988). In southern latitudes, in areas where trees are lacking, nests are frequently located in human buildings and nest-boxes (Folch 1996, Avilés & Sánchez 1997). Habitat preferences include lowland open oak *Quercus* forests, mature pine woodlands with clearings, orchards, river valleys and agricultural habitats, avoiding intensive cultivation (Cramp & Simmons 1988, BirdLife International 2010).In Europe, the species has undergone rapid declines across its global range and its breeding populations are highly fragmented (Tucker & Heath 1994, Samwald & Štumberger 1997). Therefore, it is currently classified as Near Threatened at a global level according to the IUCN Red List (IUCN 2011). Extinctions in the last century include countries such as Denmark, Finland, Sweden, Germany and Czech Republic (Glutz von Blotzheim &

Bauer 1980, Robel 1991, Samwald & Štumberger 1997, BirdLife International 2004). Exceptions to the overall European declines were reported in France and in the Spanish Extremadura (Tron *et al.* 2008, Avilés *et al.* 1999). The observed negative trends have been attributed to habitat loss due to changes in agricultural practices and persecution along the migratory routes (BirdLife International 2010). The main threats to the species that leads to negative impacts on nest-site and food availability are related to the reduction of field margins and riparian buffer strips, to conversion of grasslands to other land use and the increase use of pesticides as a consequence of agricultural intensification, to a decrease of land management of pastures due to land abandonment and to the removal of dead and damaged trees (Samwald & Štumberger 1997, BirdLife International 2010).

The Iberian Peninsula holds the most important breeding population of Rollers in Western Europe with 4.000 to 11.000 pairs. In Portugal, the population was recently estimated at 80-150 breeding pairs (Cabral *et al.* 2005) and despite the absence of quantitative data on population trends, the species disappeared from some regions and declined in others, leading to its classification as Critically Endangered in the Portuguese Red Data List (Cabral *et al.* 2005, Marques *et al.* 2005). Presently, its breeding distribution is highly fragmented and restricted to central and south areas of Portugal, especially in Beira Baixa, east Alto Alentejo and the Castro Verde region (Equipa Atlas 2008). As in south Spain, the distribution of Rollers in Portugal seem to be linked to the presence of extensive agricultural habitats, particularly cereal pseudo-steppes and open holm oak forests and by the presence of cavities in buildings (Equipa Atlas 2008). We conducted a national census in the classified Portuguese Special Protection Areas (SPAs) and Important Bird Areas (IBAs) with pseudo-steppic characteristics in the main distribution area of the species, in order to investigate species distribution, abundance and trends. The main goals of this study were: (1) to estimate the breeding population of the Roller in the Portuguese priority areas for the conservation of steppe birds, (2) to characterize and assess the availability of suitable nesting structures in each surveyed area, (3) to describe nesting preferences of Rollers, (4) to compare past and present distribution and abundance of Rollers and assess its population trend and (5) to provide management recommendations for conservation.

METHODS

» Census of the Roller breeding population

The survey included sixteen SPAs and IBAs with pseudo-steppic characteristics in Alentejo and Beira Baixa (Figure 1). In each SPA/IBA, the overall area to be prospected included open/agricultural areas selected using the Corine Land Cover 2000 map. All human made structures (houses, farm sheds, churches, mills, ruins, etc) were selected from military maps at 1:25,000 scale. We further visited all building structures within a 500 m buffer beyond the open landscape given by Corine Land Cover, coinciding mainly with forest habitats. Building structures not reported in the military maps (e.g. recent ones) but detected during the fieldwork were also visited. To increase the likelihood of species detection, every structure was visited twice: the first visit took place between 15 and 31 May and the second one between 1 and 15 June. Besides visiting the building structures, all potential perches along the pathways (fences, power and telephone lines) were checked for the presence of Rollers.

» Characterization and selection of breeding sites

All building structures were characterized according to the following variables: structure type (house, farm shed, ruin, etc), construction material (adobe, stone, brick, other), human use (inhabited or uninhabited). Principal and secondary main habitat within a 400 m radius around the structure was also recorded. Structures with cavities were classified as excellent or good (if predominantly surrounded by fallow fields or other habitat, respectively), while in the absence of cavities they were classified as medium or bad (again, if predominantly surrounded by fallow fields or other habitat, respectively). If the presence of a Roller was confirmed, the number of individuals, behaviour and/or the nest location was registered whenever possible.

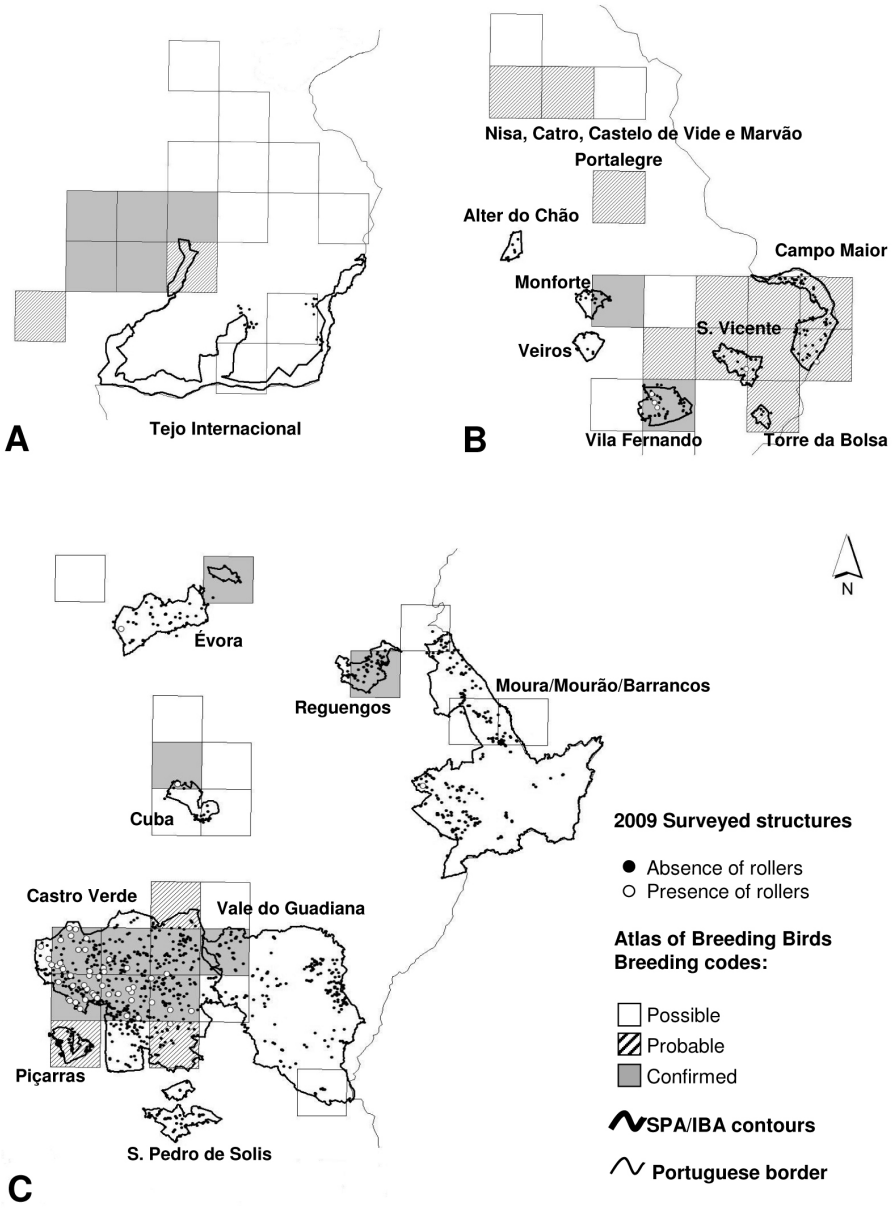


Figure 1. Location of the sixteen IBA/SPAs surveyed in 2009 for the presence of Rollers in (A) Castelo Branco, (B) Portalegre and (C) Évora and Beja districts. In each area, visited structures with (white circles) and without (black circles) breeding pairs are shown. Data on possible, probable and confirmed occurrence from the last Breeding Atlas (Equipa Atlas 2008) is provided for comparison.

Figura 1. *Localização das dezasseis IBA/ZPEs incluídas no censo de Rolieiro em 2009 realizado nos distritos de (A) Castelo Branco, (B) Portalegre e (C) Évora e Beja. Em cada área, são representadas todas as estruturas onde se confirmou (círculos brancos) ou não (círculos pretos) a nidificação de Rolieiro. Os resultados do último Atlas das Aves Nidificantes em Portugal (Equipa Atlas 2008) com informação para cada tétrada (nidificação possível, provável ou confirmada) foram adicionados para comparação com o presente censo.*

Selection of building structures by rollers, regarding the variables construction material, human use and main surrounding habitat, was analysed using the Savage selectivity index $w_i = U_i/p_i$ where U_i is the proportion of observations of rollers in a given structure and p_i is the proportion of structures with that characteristic against total available structures. This index varies from 0 (maximum negative selection) to ∞ (maximum positive selection), 1 indicating no selection. The statistical significance of this index was tested by comparing the statistics $(w_i - 1)^2/\mathrm{se}(w_i)^2$ with the corresponding critical value of a chi-squared distribution with one degree of freedom (Manly *et al.* 2003). The standard error of the index $[\mathrm{se}(w_i)]$ was calculated by $\sqrt{[(1-p_i)/(u_+^* p_i)]}$, u_+ being the total number observations (structures with rollers) sampled. Statistical significance was obtained after applying the Bonferroni correction for the number of statistical tests.

» Past and present distribution and abundance of Rollers

To evaluate potential changes in the breeding distribution, in the relative importance of each surveyed area and to assess population trends, we compared our results with existing data published in previous studies (e.g. Henriques *et al.* 2004, Silva *et al.* 2004, Marques *et al.* 2005), in two national atlas of breeding birds (Rufino 1989, Equipa Atlas 2008) and with anecdotal evidence provided by the authors and other birdwatchers. We divided the distribution area in three main regions, Castelo Branco district, Portalegre district and Évora and Beja districts, as those areas have been suggested to be the most important for Rollers (Marques *et al.* 2005, Equipa Atlas 2008).

RESULTS

» Census of the Roller breeding population

The Roller breeding population in the sixteen surveyed areas was estimated at 57 to 62 pairs. The species was detected in only six IBA/SPAs (out of the 16) and between 88.7% and 91.2% (52-55) of the breeding pairs were concentrated in the Castro Verde SPA (Table 1). Vila Fernando held 2-3 pairs, Évora, S. Vicente and Cuba held one pair each, and in Moura-Mourão-Barrancos the species was observed but breeding was not confirmed (Table 1).

» Characterization and selection of breeding sites

A total area of ca. 320700 hectares comprising 1570 building structures was surveyed in the sixteen selected IBA/SPAs (Table 1). Most of the visited structures (57%) were farms, scattered in the rural areas, often including several buildings. Ruins and isolated farm sheds represented 25% and 11% of the surveyed structures, respectively.

Around 40% of all structures were, at least, partially adobe-built (Table 1). Adobe, consisting of mud mixed with lime, was the traditional construction material in several areas and is present in more than 50% of the structures in Alter do Chão, Castro Verde, Vale do Guadiana and Cuba (Table 1). Although stone-built structures were less common compared to adobe and brick-built ones, stone was the main building material used in several areas (see Table 1). Uninhabited structures were slightly more numerous than inhabited ones (54% and 46%, respectively); Castro Verde and Vale do Guadiana SPAs hold the largest number of uninhabited structures (342 and 128, respectively, Table 1), with a high proportion of abandoned and degraded buildings.

Presence of cavities was detected in 33% of all structures, mostly in uninhabited buildings (49% and 18% in uninhabited and inhabited structures). Cavities were mainly found in stone and adobe-built structures, being almost absent from brick-built ones (55%, 38% and 12% of stone, adobe and brick-built structures, respectively). Structure suitability for Rollers (see methods) was classified as excellent in 17%, good in 19%, medium in 22% and bad in 44% of all evaluated structures (Table 1). Most structures classified as "excellent" are located in the Castro Verde SPA (n = 104), followed by Moura-Mourão-Barrancos (n = 37), S.Vicente (n = 18) and Évora (n = 15).

Table 1. Number of Roller breeding pairs and characteristics of visited building structures (material of construction, presence of cavities, use and suitability for Rollers) in the sixteen pseudo-steppic IBA/SPAs surveyed in 2009. Values correspond to the proportion of structures in each class; the sum can sometimes not reach 1 due to non-characterized structures. The total number of buildings surveyed per area is presented in brackets close to the density values.

Tabela 1. *Número de casais de Rolieiro e características das estruturas visitadas (material de construção, presença de cavidades, uso e qualidade do habitat) nas dezasseis IBA/ZPEs com habitat estepário censadas em 2009. Os valores apresentados dizem respeito à proporção de estruturas em cada classe; a soma poderá não corresponder a 1 uma vez que algumas estruturas não foram caracterizadas. O número total de estruturas visitadas é apresentada entre parêntesis junto aos valores de densidades.*

	Roller number of pairs		Area (ha)	Density of visited building structures	Construction			
	min	**max**			**adobe**	**stone**	**brick**	**other**
Tejo Internacional	0	0	25775	0.20 (51)	0	0.84	0.16	0
Alter do Chão	0	0	1318	1.06 (14)	0.64	0.07	0.29	0
Veiros	0	0	1959	0.71 (14)	0	0.79	0.21	0
Monforte	0	0	1887	1.06 (20)	0	0.40	0.60	0
Vila Fernando	2	3	5260	0.89 (47)	0.02	0.72	0.26	0
S. Vicente	1	1	3565	0.93 (33)	0	0.64	0.33	0
Torre da Bolsa	0	0	869	0.58 (5)	0	1.00	0	0
Campo Maior	0	0	9576	0.78 (75)	0.01	0.08	0.44	0.47
Évora	1	1	14707	0.54 (80)	0.01	0.73	0.23	0.04
Reguengos	0	0	6043	0.83 (50)	0.20	0.22	0.54	0.02
Moura/Mourão/ Barrancos	0	1	80565	0.31 (250)	0.46	0.16	0.37	0
Cuba	1	1	4081	0.74 (30)	0.53	0	0.47	0
Vale do Guadiana	0	0	76578	0.28 (211)	0.75	0.07	0.18	0
Castro Verde	52	55	79066	0.68 (535)	0.67	0.07	0.22	0
Piçarras	0	0	2827	3.08 (87)	0.29	0.16	0.55	0
S. Pedro de Solis	0	0	6610	1.03 (68)	0.09	0.53	0.38	0
TOTAL	**57**	**62**	**305728**	**0.51 (1570)**	**0.44 (698)**	**0.22 (341)**	**0.30 (466)**	**0.03 (41)**

	Presence of cavities		Inhabited?		Structure suitability			
	yes	**no**	**yes**	**no**	**bad**	**medium**	**good**	**excellent**
Tejo Internacional	0.55	0.45	0.22	0.78	0.04	0.41	0.27	0.27
Alter do Chão	0.29	0.71	0.21	0.79	0.29	0.43	0.29	0
Veiros	0.71	0.29	0.21	0.79	0.14	0.14	0.50	0.21
Monforte	0.40	0.60	0.30	0.70	0.40	0.60	0	0
Vila Fernando	0.60	0.40	0.40	0.60	0.15	0.34	0.26	0.26
S. Vicente	0.73	0.24	0.48	0.48	0.03	0.18	0.21	0.55
Torre da Bolsa	0.80	0.20	0.20	0.80	0.20	0	0.40	0.40
Campo Maior	0.43	0.57	0.48	0.52	0.41	0.16	0.35	0.08
Évora	0.41	0.59	0.71	0.29	0.25	0.13	0.19	0.19
Reguengos	0.18	0.82	0.58	0.42	0.72	0.08	0.14	0.04
Moura/Mourão/ Barrancos	0.36	0.63	0.63	0.36	0.38	0.13	0.19	0.15
Cuba	0.53	0.33	0.47	0.53	0.07	0.27	0.10	0.43
Vale do Guadiana	0.13	0.87	0.42	0.58	0.42	0.04	0.01	0.01
Castro Verde	0.33	0.59	0.34	0.64	0.29	0.27	0.15	0.19
Piçarras	0.13	0.87	0.63	0.37	0.87	0	0.11	0.01
S. Pedro de Solis	0.32	0.68	0.43	0.57	0.40	0.26	0.26	0.07
TOTAL	**0.33 (524)**	**0.64 (997)**	**0.45 (706)**	**0.54 (849)**	**0.35 (553)**	**0.19 (300)**	**0.16 (255)**	**0.15 (234)**

Rollers positively selected building structures surrounded by fallows ($w_i = 1.93$, $p<0.01$) and avoided structures surrounded by "montado" and other habitats, such as irrigated cultures, vineyards, olive plantations and orchards ($w_i = 0.07$, $p<0.01$ and $w_i = 0.0$, $p<0.01$, respectively, Table 2 and Figure 2). Rollers used significantly more than expected the adobe-built structures ($w_i = 1.61$, $p<0.05$), while no preference was shown when comparing uninhabited and inhabited structures (Table 2 and Figure 2).

Although most pairs nested in natural cavities, around 40% of the population occupied artificial nests (mostly in the Castro Verde SPA, Figure 3). Artificial nests, massively provided in the last years to recover the Lesser Kestrel breeding population (see Catry *et al.* 2009), included built-cavities in adobe and concrete walls, wooden nest-boxes and clay pots. The relatively high percentage of breeding sites in brick-built structures (15%) is due to a high proportion (eight out of nine) of pairs nesting in artificial nests. Most building structures occupied by Rollers held a single pair; three structures had two breeding pairs and in another three we estimated one to two pairs. The presence of Rollers was frequently associated with structures holding Lesser Kestrel colonies (57% of the cases) and other cavity-nesters such as the Little (*Athene noctua*) and Barn (*Tyto alba*) Owls and the Common Kestrel (*Falco tinnunculus*).

DISCUSSION

» Distribution, abundance and population trends of the Roller in Portugal

There is some evidence that the Roller was not a very common species in the beginning of the 20th century (Marques *et al.* 2005). The first national atlas of breeding birds refers its distribution along the eastern part of the country (Rufino 1989) but recent data suggest its extinction in areas such as Trás-os-Montes and Beira Alta (Equipa Atlas 2008, L. Reino, pers. comm.). The distribution of the Roller became fragmented and restricted to inland areas in the central and southern regions of the country; the three most important regions include Castelo Branco, Portalegre and Beja districts (Marques *et al.* 2005, Equipa Atlas 2008).

» Castelo Branco district

Although we have found no Rollers in the Tejo International SPA, the Castelo Branco/Idanha-a-Nova region might still be a relatively important area for the species (Equipa Atlas 2008). According to the last atlas of breeding birds (Equipa Atlas 2008), the reproduction of this species was confirmed in five 10 x 10 UTM cells, all outside the protected area (Figure 1). This region is characterized by a high diversity of habitats, which include steppe habitat but also Holm oaks and Rollers are known to use both cavities in human structures and trees (S. Infante, pers.comm.; Marques *et al.* 2005). Although there is no quantitative data on the population abundance, the species is probably rare and declining (C. Pacheco and S. Infante, pers. comm.).

» Portalegre district

Outside the surveyed SPA/IBAs, the species used to be relatively common, particularly around Arronches, Castelo de Vide, Marvão, Nisa and Portalegre (G. Elias and L. Venâncio, pers.comm., Rufino 1989, Elias 1995). Nevertheless, the population is thought to be declining, as suggested by recent observations (G. Elias and L. Venâncio, pers. comm.) and by the absence of confirmed breeding in the last atlas survey (Figure 1, Equipa Atlas 2008). Among the surveyed areas, there is no information on the past occurrence of Rollers in either the Alter do Chão IBA and Veiros SPAs, where no individuals were observed during the present study. In the Monforte SPA, no Rollers were found; according to the breeding atlas (Equipa Atlas 2008) one breeding pair was detected outside this SPA (Figure 1).

In Vila Fernando, the Roller breeding population was estimated in two to three pairs, showing a decline from 2003, when five to six pairs were recorded in the area (Silva *et al.* 2004). In S. Vicente, two breeding pairs bred in the area in 2007 but a single one was confirmed in 2009, occupying a nest-box provided in the same year (J.P. Silva, unpubl. data). Rollers might have been common in the Campo Maior SPA, but in 1997 there were only two pairs left (J.P. Silva, unpubl. data). In the present survey, the species was only detected using an old structure in Spain, next to the Portuguese border, agreeing with the lack of a breeding record during the breeding atlas surveys (Figure 1). Elias (1995) described the road from Elvas to Torre da Bolsa as one of the best places in Portugal to observe the

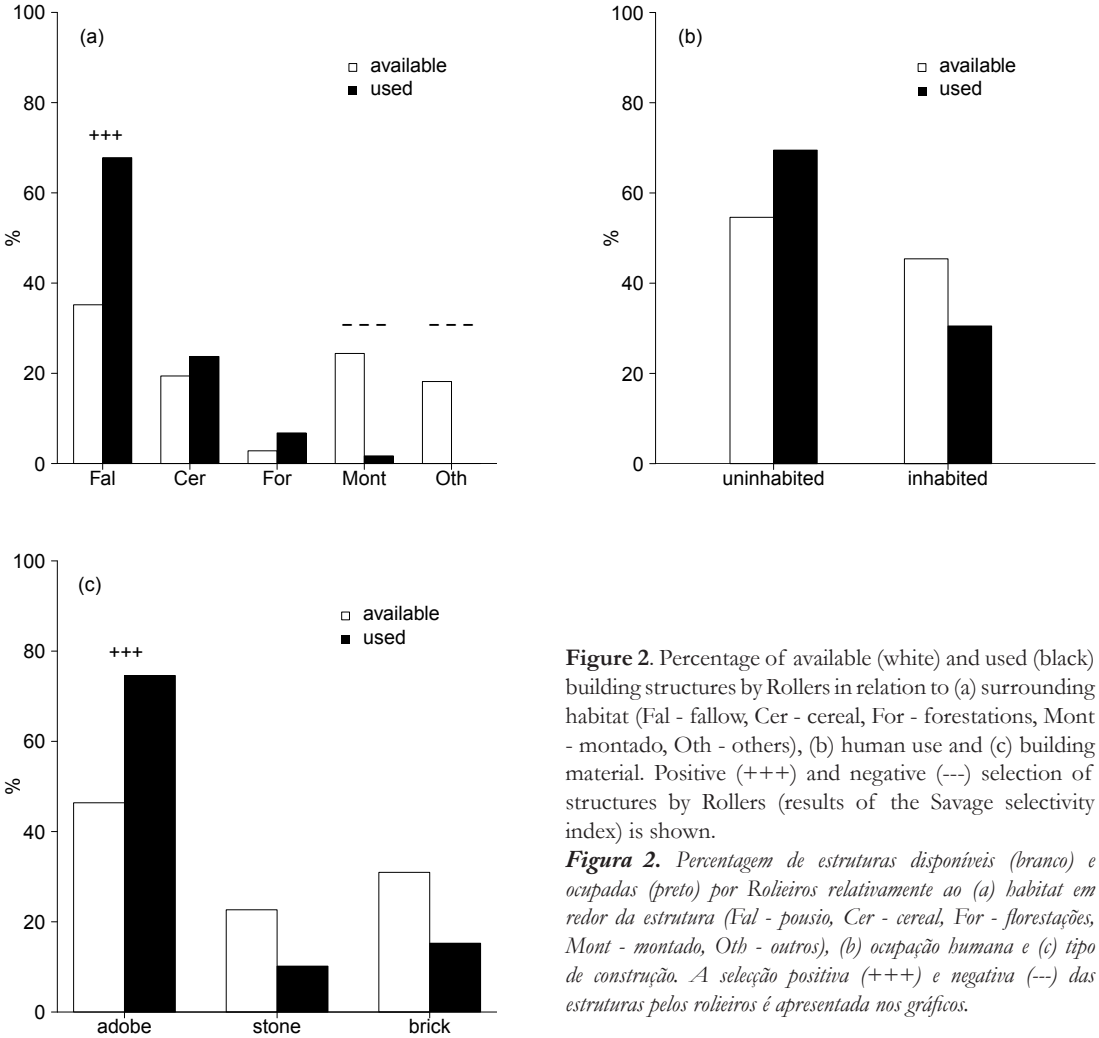


Figure 2. Percentage of available (white) and used (black) building structures by Rollers in relation to (a) surrounding habitat (Fal - fallow, Cer - cereal, For - forestations, Mont - montado, Oth - others), (b) human use and (c) building material. Positive (+++) and negative (---) selection of structures by Rollers (results of the Savage selectivity index) is shown.

Figura 2. *Percentagem de estruturas disponíveis (branco) e ocupadas (preto) por Rolieiros relativamente ao (a) habitat em redor da estrutura (Fal - pousio, Cer - cereal, For - florestações, Mont - montado, Oth - outros), (b) ocupação humana e (c) tipo de construção. A selecção positiva (+++) e negativa (---) das estruturas pelos rolieiros é apresentada nos gráficos.*

Roller. Nevertheless, the species was not found either during this and the last breeding atlas surveys, suggesting a decline of its breeding population (Figure 1).

» Évora and Beja districts

Despite the existence of one breeding record in Reguengos from the last atlas of breeding birds (Equipa Atlas 2008), no Rollers were found in the area during the present survey. In Mourão, the Roller was a regular breeder a few decades ago, and although there is no quantitative data on its past abundance, it should have been relatively common (Elias 1995, C. Cruz and G. Elias, pers. comm.). In the last decade, in Moura-Mourão-Barrancos SPA, there are only occasional observations of the species and its reproduction has not been confirmed in the last years (M. Lecoq, pers. comm.; Equipa Atlas 2008; this study), suggesting a rapid population decline. In Évora, the reproduction of the Roller has been referred in the past (Rufino 1989) but presently, its occurrence seems to be irregular (C. Cruz, unpubl. data). During the last national atlas survey only one pair was detected outside the SPA (Equipa Atlas 2008), while in the present study one pair was detected in the southern

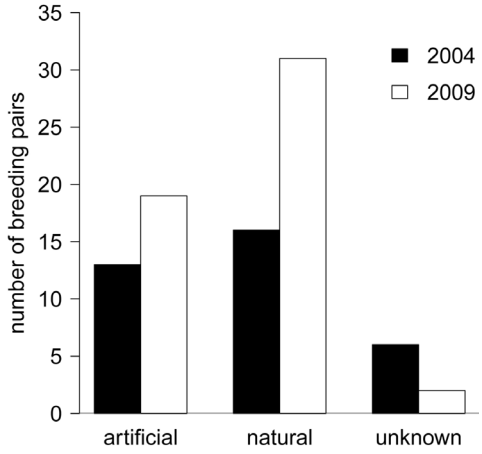
Table 2. Values of Savage selectivity index (ω^i), standard error (SE) and significance level (*P*) for the variables surrounding habitat (main habitat in a 400 m buffer around the structure) and structure occupation (building material and human use) in the sixteen pseudo-steppic IBA/SPAs surveyed in 2009.

Tabela 2. *Valores do Índice de Savage (ω^i), erro padrão (SE) e nível de significância (P) para as variáveis habitat em redor da estrutura (habitat dominante num raio de 400 m em volta da estrutura) e ocupação da estrutura (tipo de construcção e ocupação humana) nas dezasseis IBA/ZPEs com características pseudo-estepárias censadas em 2009.*

	ω^i	SE	*P*
Habitat			
fallow	1.93	0.18	< 0.01
cereal	1.22	0.27	n.s.
forestation	2.40	0.76	n.s.
montado	0.07	0.23	< 0.01
others	0.00	0.28	< 0.01
Building material			
adobe	1.61	0.18	< 0.05
stone	0.45	0.32	n.s.
brick	0.49	0.26	n.s.
Human use			
uninhabited	1.27	0.16	n.s.
inhabited	0.67	0.19	n.s.

Figure 3. Comparison of the number of Roller breeding pairs in the Castro Verde SPA in 2004 (Henriques *et al.* 2004) and 2009 (present study). Artificial nests include built cavities in adobe and concrete walls, wooden nest-boxes and clay pots. Natural nests are mainly non-built cavities.

Figura 3. *Ocupação de ninhos artificiais e naturais por Rolieiro na ZPE de Castro Verde em 2004 (Henriques* et al. *2004) e 2009 (presente estudo). Os ninhos artificias incluem cavidades em taipa e cimento, caixas-ninho e potes de barro. Os restantes ninhos naturais localizam-se maioritariamente em cavidades naturais.*



part of this area (Figure 1). The absence of previous observations of Rollers in the Cuba region (Rufino 1989) suggests a recent (post-1980) colonization of the area. Nonetheless, the species might be already declining, as two to three pairs were known to breed within the surveyed area between 2003 and 2007 (I. Catry unpubl. data, Pacheco 2009), whereas a single pair was detected in the present survey. As in other areas, the Roller was a regular and frequent breeder a few decades ago in Vale do Guadiana SPA, (Rufino 1989, Elias 1995, Rocha 1999). In the present survey, the species was not detected in the entire area, suggesting a recent decline (last decade) and its possible local extinction. The Castro Verde SPA is the only area where there was a clear increase in the Roller breeding population. According to testimonies from local inhabitants, the Roller was absent or very rare in the area a few decades ago and its increase is thought to be related with an increase in nest-site availability in degraded structures due to agricultural abandonment and desertification (Marques *et al.* 2005). The number of breeding pairs in Castro Verde SPA increased from 35-40 in 2004 (Henriques *et al.* 2004) to 52-55 in 2009. The last survey included, however, an area of ca. 6,000 ha (corresponding to the enlargement of the SPA in 2008), not surveyed in 2004 and where six pairs were found. No Rollers were detected in either Piçarras or S. Pedro de Solis SPAs. In the first area, one pair has been recorded in 2002 in one of the surveyed buildings (R. Tomé, pers. comm.), while there are no data on the occurrence of Rollers in the second one (Figure 1).

Overall, the results of this study confirm the scarcity and highly fragmented distribution of the Roller in Portugal, with residual breeding populations in all SPAs where the species was detected, apart from Castro Verde. Despite the

lack of quantitative data, the evidence provided in this study suggests a general negative trend of the Portuguese population, especially in Alto Alentejo and Beira Baixa regions. Only in Castro Verde SPA the species is relatively abundant (52-55 pairs), confirming this area as the most important for the species, as it was previously suggested (Marques *et al.* 2005, Equipa Atlas 2008).

» Selection of breeding sites

Rollers are obligate hole-nesters that do not excavate their own cavities (Cramp & Simmons 1988). In open areas, where trees are scarce, breeding pairs are highly dependent on human constructions (Folch 1996, Gragera *et al.* 2000). A previous study in southwest Spain showed a high preference for inhabited buildings, and suggested a lower predation pressure in these places as a potential explanation (Gragera *et al.* 2000). In the surveyed areas of this study, most breeding pairs were found in uninhabited structures, probably reflecting their higher availability (55% vs. 45%) as no preference towards uninhabited/inhabited structures was found (Figure 2, Table 2). The high availability of cavities and fissures in adobe-built structures (created by the wind and rain and lack of maintenance) might explain the preference of this type of building material by breeding rollers. Recently built houses tend to be brick-built and are less likely to provide nest-sites to rollers. The positive selection of structures surrounded by fallows reflects the higher density and accessibility of arthropods (the main diet of Rollers) in this habitat (e.g. Gragera *et al.* 2000, Avilés & Parejo 2004, Poole & Tron 2007).

Almost 90% of the total population was found in Castro Verde SPA, where 20% of all structures surveyed (n=104) were classified as "excellent" for the species (with cavities and mainly surrounded by fallows). Although Castro Verde had the highest number of visited structures (n = 535), the density of structures was not amongst the highest values recorded. Besides Castro Verde, all breeding Rollers were recorded in five of the six areas with the highest number of "excellent" structures (range 12-37). In previous studies, relatively high densities of breeding pairs have been found in areas with high nest-site availability and with plentiful food supply (Cramp & Simmons 1988, García & Guzmán 1991, Butler 2001, Tidmarsh & Di Corrado 2003). We should stress that cavity suitability for Rollers was not assessed during the survey.

» Threats, future scenarios and management implications

Population declines of the Roller across most of the European range have been often attributed to either loss or degradation of foraging or nesting habitat (BirdLife International 2004). In Portugal, most of the surveyed areas in this study suffered large habitat losses in the last decades, showing declines in the extent of fallow land (positively selected by rollers in our study), cereal and legume fields, and increases of unsuitable habitats, such as irrigated crops (Pinto *et al.* 2005, Costa *et al.* 2003). Moreover, the characterization of building structures suggests that the presence of Rollers may be strongly constrained by the lack of suitable cavities (and adobe-built structures) in most areas. The presence and abundance of the species in the Castro Verde area, characterized by very extensive agricultural practices and low densities of trees, is probably the result of two main driving factors: 1) the implementation, in 1995, of the agri-environmental measure "Castro Verde Zonal Plan", that aims to maintain favourable feeding habitat for a range of steppe-bird species (Moreira *et al.* 2007) and 2) the high availability of adobe-built structures and the massive provisioning of artificial nest-sites. Local increases in breeding populations due to nest-box provisioning have also been recorded in the steppe areas of southwest Spain, contrasting with the general negative population trend recorded in the country (Avilés & Sánchez 1997, Marti & Del Moral 2004). If the breeding population in Castro Verde continues to increase, this area could act as a source to enhance breeding populations in other areas. Thus, it is crucial to maintain and restore the remaining Portuguese pseudo-steppe areas, avoiding agriculture intensification and promoting the traditional extensive practices. The maintenance of fallow and uncultivated strips could ensure the existence of healthy and diverse arthropod populations while the provisioning of artificial nest-sites could tackle the lack of natural ones.

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REFERENCES

Avilés, J.M. & A. Sánchez 1997. Evolución del número de parejas reproductoras de Carraca Coracias garrulus en cinco hábitats de Extremadura. *Butlletí del Grup Català d'Anellament* 14: 25-29.

Avilés, J.M., J.M. Sánchez, A. Sánchez & D. Parejo 1999. Breeding biology of the Roller Coracias garrulus in farming areas of the southwest Iberian Peninsula. *Bird Study* 46: 217-223.

Avilés, J.M. & D. Parejo 2004. Farming practices and Roller *Coracias garrulus* conservation in south-west Spain. *Bird Conservation International* 14: 173-181.

BirdLife International 2004. *Birds in Europe: population estimates, trends and conservation Status*. BirdLife Conservation Series No 12. BirdLife International, Cambridge.

BirdLife International 2010. Species factsheet: *Coracias garrulus*. Downloaded from http://www.birdlife.org on 16/10/2010.

Butler S. 2001. *Nest-site selection of the European Roller (*Coracias garrulus*) in the Vallée des Baux de Provence*. MSc report, University of York.

Cabral, M.J. (coord.), J. Almeida, P.R. Almeida, T. Dellinger, N. Ferrand de Almeida, M.E. Oliveira, J.M. Palmeirim, A.L. Queiroz, L. Rogado, & M. Santos-Reis (eds.) 2005. *Livro Vermelho dos Vertebrados de Portugal*. Instituto da Conservação da Natureza. Lisboa. 660 pp.

Catry, I., R. Alcazar, A.M.A. Franco & W.J. Sutherland 2009. Identifying the effectiveness and constraints of conservation interventions: A case study of the endangered lesser kestrel. *Biological Conservation* 142: 2782-2791.

Costa. L.T., Nunes, M., Geraldes, P. & Costa, H. 2003. *Zonas Importantes para as Aves em Portugal*. Sociedade Portuguesa para o Estudo das Aves, Lisboa.

Cramp, S. & K.E.L. Simmons 1988. *The birds of the Western Paleartic*, Vol.V. Oxford University Press, Oxford.

Elias, G. 1995. Onde observar o Rolieiro *Coracias garrulus*. *Pardela* 1: 7-8.

Equipa Atlas 2008. *Atlas das Aves Nidificantes em Portugal (1999-2005)*. Instituto da Conservação da Natureza e Biodiversidade, Sociedade Portuguesa para o Estudo das Aves, Parque Natural da Madeira e Secretaria Regional do ambiente e do Mar. Assírio & Alvim. Lisboa.

Folch, A. 1996. Distribución y status de la Carraca Coracias garrulus en Cataluña. Pp 113-119 *in* J. Fernández & J. Sanz-Zuasti (Eds.). *Conservación de las aves esteparias y sus hábitats*. Junta de Castilla y Léon.

García, A.S. & J.M.S. Guzmán 1991. Resultados de ocupacion de cjas anidaderas en tendidos electricos en Extremadura (oeste de España): 1986-1990. *Ecología* 5: 375-381.

Gragera, F., J.E. Rodríguez & L. Salguero 2000. La población nidificante de Carraca en dos comarcas de Badajoz. *Quercus* 170: 10-14.

Glutz von Blotzheim, U.M. & K.M. Bauer 1980. *Handbuch der Vögel Mitteleuropas*. Akademische Verslagsgessellchaff, Wiesbaden.

Henriques, I., Marques, A. T. & Moreira, M. I. 2004. Contributo para o conhecimento e a conservação do Rolieiro (*Coracias garrulus*) na ZPE de Castro Verde. *Actas do Congresso Internacional: Ecossistemas Agrícolas e Riqueza Biológica – Principais Ameaças e Medidas de Conservação*. Liga para a Protecção da Natureza.

Lisboa.

IUCN 2011. IUCN *Red List of Threatened Species*. Version 2011.2. Downloaded from http://www.iucnredlist.org on 28 November 2011.

Manly, B.F.J., McDonald, L.L., Thomas, D.L., 1993. *Resource Selection by Animals*. Chapman and Hall, London.

Marques, A.T., I. Henriques, I. Catry & M.I. Moreira 2005. Distribution of the Roller *Coracias garrulus* in Portugal, an historical approach. *Ardeola* 52: 173-176.

Marti, R. & J.C. Del Moral (Eds.) 2004. *Atlas de las Aves Reprodutoras de Espana*. Dirección General de Conservación de Espana. Dirección General de Conservación de la Naturaleza – Sociedad Espanola de Ornitología, Madrid.

Moreira, F., Leitão, P.J., Morgado, R., Alcazar, R., Cardoso, A., Carrapato, C., Delgado, A., Geraldes, P., Gordinho, L., Henriques, I., Lecoq, M., Leitão, D., Marques, A.T., Pedroso, R., Prego, I., Reino, L., Rocha, P., Tomé, R., Osborne, P.E., 2007. Spatial distribution patterns, habitat correlates and population estimates of steppe birds in Castro Verde. *Airo* 17, 5–30.

Pacheco, C. 2009. Monitorização da avifauna no Bloco de Rega do Alvito-Pisão e área remanescente da ZPE/IBA de Cuba. EDIA. Relatório não publicado.

Pinto, M., Rocha, P. & Moreira, F. 2005. Long-term trends in great bustard (*Otis tarda*) populations in Portugal suggest concentration in single high quality area. *Biological Conservation* 124: 415-423.

Poole, T.F. & F. Tron 2007. *Assessing the selection of foraging habitats by the European Roller, Coracias garrulus (L.) in the Vallée des Baux*. Synthèse des études et travaux de conservation A Rocha France 2007.

Robel, D. 1991. The last breeding attempt of the Roller (*Coracias garrulus*) unsuccessful in Germany. *Vogelwelt* 112: 148-149.

Rocha, P.A. 1999. A reprodução do Rolieiro *Coracias garrulus* no Baixo Alentejo no período 1996-1999. Pp 19-21 *in* Beja, P., P. Catry & F. Moreira (Eds). *Actas do II Congresso de Ornitologia da Sociedade Portuguesa para o Estudo das Aves*, Sociedade Portuguesa para o Estudo das Aves, Lisboa.

Rufino, R. (Coord) 1989. *Atlas das Aves Nidificantes em Portugal Continental*. Serviço Nacional de Parques e Reservas e Conservação da Natureza, ICN, Lisboa.

Samwald, O. & B. Štumberger 1997. Roller (*Coracias garrulus*). Pp 436-437 *in* W.J.M. Hagemeijer & M.J. Blair (Eds.). *The EBCC atlas of European breeding birds: Their distribution and abundance*. T & AD Poyser, London.

Silva, J.P., Leitão, D., Santos, E., Moreira, F., Prego, I., Pinto, M., Lecoq, M., Catry, T., Pedroso, R. 2006. Preliminary results of the Little Bustard's census in Alentejo (Portugal), in: Leitão, D., Jolivet, C., Rodriguez, M., Tavares, J. (Eds.), Bustard conservation in Europe in the last 15 years: current trends, best practice and future priorities. RSPB/BirdLife, Bedfordshire, pp. 55-59.

Silva, P., L.T. Costa & J.P. Silva 2004. Caracterização da avifauna estepária e suas relações com o uso do solo na Zona Importante para Aves de Vila Fernando/Veiros. *Airo* 14: 41-53.

Tidmarsh, R. & C. Di Corrado 2003. *Hunting success as a function of habitat and perch type in the European Roller (*Coracias garrulus*) population of the Valley des Baux in 2003*. Synthèse des études et travaux de conservation A Rocha France 2003.

Tron, F., A. Zenasni, G. Bousquet, P. Cramm & A. Besnard 2008. Réévaluation du statut du Rollier d'Europe *Coracias garrulus* en France. *Ornithos* 15-2: 84-89.

Tucker, G.M. & M.F. Heath 1994. Birds in Europe: their conservation status. BirdLife International, Cambridge.

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